Apartment 50: Paper 009

AuthorityEffect: None. This is fictional examination staging, not a historical quotation or a source of mathematical authority.

Landau points at the declared carrier. Woland asks which claim its checker can establish. Behemoth asks for the smallest valid near miss. Their task in this companion is **read a claim at the checker scope**.

Begin with the study guide, then attempt the quiz before reading the separate instructor key. A lower-case wire token names a role only after its interpretation is declared. Arrows sketch graph edges; they do not automatically attach higher-dimensional cells. A receipt records its stated finite check and grants no authority to extend Genesis or make a physical claim.

The examiner's standard is a concrete calculation, an explicit counterexample, or a source-bound missing obligation. The manuscript source reading governs the answer.

Study guide: Paper 009

AuthorityEffect: None. The Apartment 50 preface is fictional framing. Attempt the quiz before reading the instructor key. No learner study or qualification is reported.

Route through the paper

Level	Read	Practice	Move on when you can
1	Section 1	Reduce a broad claim	Separate assumption, witness, execution, and obligation
2	Section 3 and notebook	Predict both operations	Explain the invariant and identify changed input
3	Section 4	Corrupt controls separately	Distinguish feasibility, closure, objective matching, and digest binding
4	Section 4.1 and trace	Reconstruct the four cells	Give numerators, denominators, and label provenance
5	Sections 5–6 and Lean	Read definitions before names	Distinguish formal facts, census scope, and runtime correspondence
6	Appendix A.1–A.4	Follow each verdict to its operation	Identify four families and five probes
7	Appendix A.5	Inspect corrections	Separate necessary evidence from guarantees about a checker
8	Sections 6–8	Write a scoped record	Preserve refusals, missing bridges, and unexecuted dependencies

Prepare the exact calculation

Use edges (0,1), (1,2), (2,0). Write all three endpoint differences before subtracting. Each potential occurs once with each sign around the cycle; the differences sum to zero. This cancellation explains every potential on the declared cycle. A graph path is not an automatically filled simplex.

Replacing claims by df changes the input. Subtracting df from fixed claims preserves their cycle sum. Save original and transformed inputs: a low score on replacement data does not show the original obstruction disappeared.

Predict the displayed numbers, then run:

```
python tools/replay_notebook.py
```

The default executes all five ordinary Python cells in a fresh CPython process, checks assertions, and compares stored outputs without rewriting the notebook. Jupyter kernel execution is a separate `--engine jupyter` check. Neither mode runs the native graph checker or grades a learner's explanation.

Prepare the historical recount

Read evidence provenance. "Refused" and "not refused" mean the original checker's decisions, not physical ground truth. Separate CHALLENGE and CONCUR, check all cells total ten, and keep a later checker decision as a distinct observation.

False-positive fraction divides challenges of not-refused proposals by all not-refused proposals. Precision divides challenged refused proposals by all challenges. Different denominators answer different questions. Use exact fractions for the new recount; zero denominators are undefined.

```
python tools/review_capstone.py
python -m pytest -q
```

The capstone verifies retained digests and the complete condition census, changes one verdict in a copy, and ignores the constant caught flag. Tests also reject non-Boolean labels, unknown verdicts, and label/status disagreement. The gate record inspection does not execute the gate. Complete the written report too.

Diagnose a missed answer

Mistake	Return to	Correction
A hash or token is treated as proof	Level 1	State the missing property check
Subtraction is confused with replacement	Level 2	Track original input, operation, residual, and sum

Mistake	Return to	Correction
Equal LP objectives replace feasibility	Level 3	Check both feasible regions
The caught flag yields perfect detection	Level 4	Count actual decisions against historical labels
A theorem name certifies a runtime	Level 5	Identify the correspondence obligation
Four families are called four records	Level 6	Count F2a and F2b separately
A bound receipt makes a critique true	Level 7	Separate identity, authentication, and validity
A retained report is called a fresh replay	Level 8	Record the command actually run

Write a review record

Include claim text, source revision, artifact path/digest, assumptions, exact input, command, environment or executable pin, observed result, review disposition, and remaining obligation. Mark a command `NOT_RUN` until observed. The manuscript's dispositions are review terms, not a new runtime API.

The v0.1 review record separates fresh execution from earlier reports. Deposit and ZIP checks establish file custody, not the truth of every answer. A pinned Lean build is a separate dependency check.

Instructor key: Paper 009

Use after attempting the quiz. Equivalent scoped answers count. The local rubric awards result, reason, and scope for each question. **AuthorityEffect: None.** This key provides practice feedback, not a credential.

Level 1

1. Example: on the declared triangle, check whether replacement (5,0,0) of original (1,1,1) is refused as `DefectIncrease` and active claims remain unchanged. Name the native-backed checker, score definition, revision, command, and stored state. This is a proposed check until executed.
2. Assumption: declared graph and allowed potential move. Witness: proposed potential. Receipt: observed execution record. Obligation: missing native replay. All except an observed receipt may precede execution.
3. `qed` is an assertion; a matching hash identifies declared bytes relative to the digest. The theorem needs proof/check under stated hypotheses. Custody does not authenticate the producer or establish truth.

Level 2

4. df=(1/3,1/3,−2/3), h−df=(0,0,1). Original and residual sums are both 1.
5. (f1−f0)+(f2−f1)+(f0−f2)=0. Constant shifts cancel on every edge. This argument covers all potentials on the declared cycle.
6. df=(2,−1,−1), h−df=(0,0,5), original/residual sum 5. Replacement by df has sum 0 and changes input; it does not remove the original sum.

Level 3

7. Both witnesses must be feasible. Positive: feasible matching witnesses. Separate negatives: violate a primal bound; introduce nonzero dual vertex flow; retain feasibility but choose a different objective. Proposed or source-inspected controls are not fresh results until executed.
8. Recomputed digest from changed bytes should fail comparison with the old key. String length checks only shape; neither test establishes content truth.
9. No: always-pass accepts the valid case too. Pair a valid bound node with mutated content retaining its old key. Expect pass/failure respectively. The distinction is byte binding under declared serialization.

Level 4

10. TP=6, FN=0, FP=2, TN=2, with the positive conventions in the question. Refused row 6, not-refused row 4, total 10.
11. Challenged/refused=6/6=1; challenged/not-refused=2/4=1/2; refused/challenged=6/8=3/4; label agreement=(6+2)/10=4/5. These

describe these records under these historical labels.

12. The caught flag is also true for both concurrences. The retained later checker replay refuses all ten; the four obstruction claims lack checked witnesses. Earlier "sound" meant only not refused at that time. No physical ground truth or general rate follows. No adversary blocks anything the checker would otherwise accept. The later receipt does not pin its checker revision, and reading it does not freshly replay it.

Level 5

13. The named report lists 32 theorems (28 axiom-free, four with dependencies); the full report includes 516 module declarations, including generated and private ones. Inspection does not compile Lean. Fresh compilation must record the pinned toolchain and actual commands.
14. Equal self-hashes fail an unequal-hash predicate. Distinct hashes do not prove statistical independence or independent error sources. Runtime correspondence is a separate obligation.
15. Scores need independent computation from original inputs; flags need provenance and runtime correspondence. Supplied-value routing is a narrow fixture test, not comprehensive semantic verification.

Level 6

16. Act IV replaces the cochain with zeros; Act II scores those zeros. The original sum 3 cannot be removed by potential subtraction. This is a constructed failure at a historical commit, not its measured frequency.
17. F2a misses a counterexample without keywords; F2b blocks a negated keyword. The pair isolates lexical adjudication. Five probes, four defect families.
18. F3's run-key hash does not identify stored evidence, and constant flags do not check balance or norm bounds. F4's chain binds unsupported template text, including unexecuted repairs. Consistent bytes do not verify claims.

Level 7

19. Necessary conditions: eligible candidate, no applicable blockers, and an accepted checker receipt bound to the candidate. A declared carrier also requires verified exact joint zeros. With no declared carrier, that zero requirement does not apply. A supplied zero alone is insufficient. This describes the pinned correction's source, not a fresh gate execution.
20. Binding identifies bytes; authentication identifies a producer; correctness establishes the checker implements the claimed property; critique validity requires a checked argument. The latter do not follow from digest equality or successful typed parsing.
21. "I inspected the pinned correction and read the historical reports of 66 and 44 passing tests; I did not execute them." A fresh claim needs the checkout,

executable/dependency pins, named test commands, dated outcomes, and preserved outputs. Historical JSON inspection is a different operation.

Level 8

22. (a) Supported within scope; (b) Refused claim; (c) Open obligation; (b) Not run. Different claims can have different dispositions.
23. Claim, revision, path/digest, assumptions, exact input, command, environment or executable pin, result, disposition, obligation. RFC 8785 also requires primitive serialization and other conditions. A zero-denominator fraction is undefined, represented by `None`, not zero or perfect performance.
24. Record unchanged residual, attempted move family, and invariant. New representations need new assumptions and checks. A trigger does not license protected input changes or demonstrate an effective search escape.

Capstone key (8 points)

1. Three retained digests match provenance. This establishes custody relative to the declarations, not independent authentication of the execution.
2. Exactly ten distinct conditions: two named lattices, values $-4,-2,0,2,4$. The row count excludes duplicates hidden by a set.
3. Cells (6,0,2,2); fractions 1,1/2,3/4,4/5 with Question 11 meanings.
4. One refused challenge changed to concurrence gives (5,1,2,2), and 5/6 in the refused row. The copied control is not a new model trial.
5. Changing only the caught flag leaves cells unchanged.
6. F1 input replacement; F2a/F2b keyword adjudication; F3 phantom references; F4 templated receipts. Five records, four families.
7. Preserve original files and identify every altered copy as a control.
8. Example: "At the provenance-bound inputs, `python tools/review_capstone.py` passed custody, census, and recount controls, yielding (6,0,2,2). This freshly recounts retained data; no model, native checker, or gate was rerun. A native replay still needs the checker revision and executable." Add your actual environment, date, and artifact identities. The tool does not grade prose.

Theoretical minimum: Paper 009

Read the fictional Apartment 50 preface. This is a local practice quiz, not a historical Landau examination or a qualification. **AuthorityEffect: None.** Use the study guide for preparation and the separate instructor key after attempting the questions. Cite a section, definition, input, or artifact for each answer.

The triangle has edges (0,1), (1,2), (2,0), in that order. Potentials live on vertices; edge differences are target minus source. Use integers and exact fractions for new calculations. Historical records retain their representation.

Level 1 — Claim, witness, receipt (Section 1)

1. Rewrite "the system verifies repairs" as one inspectable claim. Name the input, allowed operation, checker, and expected stored state.
2. Distinguish an assumption, witness, receipt, and open obligation with a triangle example. Which can be supplied before execution?
3. A model prints `qed` and an author supplies a correctly hashed output file. What has been established? What check is still needed for the theorem claim?

Level 2 — Input preservation (Section 3)

4. For h=(1/3,1/3,1/3) and f=(0,1/3,2/3), calculate df, h−df, and both cycle sums.
5. Explain why every vertex potential has zero cycle sum in its edge differences. What happens if a constant is added to all vertex potentials?
6. For h=(2,−1,4) and f=(0,2,1), compute h−df. Compare subtraction with replacement of h by df. Which operation changes the declared claim input?

Level 3 — Controls that distinguish cases (Section 4)

7. An LP checker accepts whenever primal and dual objectives match. State the missing conditions. Design separate failures for a primal bound, dual closure, and a feasible but nonmatching objective.
8. Node content changes while its old digest remains. Compare digest recomputation with a check that the digest string has 64 characters.
9. A checker passes a valid example. Does this rule out always-pass behavior? Give a positive/negative pair and name the tested distinction.

Level 4 — Recount the historical bout (Section 4.1)

10. Reconstruct the four cells from six refused proposals (all challenged) and four not-refused proposals (two challenged, two concurred). CHALLENGE is the positive decision; historical refusal is the positive label.
11. Compute each row's challenge fraction, the refused fraction among challenges, and overall label agreement. Give denominators.

12. Why does the caught flag being true in all ten rows not show perfect detection? Explain the later recorded checker replay, and what these trials do not establish about physical ground truth or adversary benefit.

Level 5 — Formal scope and supplied inputs (Sections 5–6)

13. Explain 32 named theorems versus 516 environment declarations. Does inspecting their reports freshly compile Lean?
14. A Lean predicate calls unequal parameter hashes "independent." What does its self-comparison theorem establish, and which independence claim remains open?
15. A toy gate receives caller-supplied zero scores and clean flags and returns `PASS_TO_MASTER`. Name two missing bridges before claiming semantic verification.

Level 6 — Audit the historical gate (Appendix A)

16. F1 starts from (1,1,1), then a lift returns zeros and the gate authorizes. Locate the input change. Is this a potential repair?
17. Explain the paired F2 failures: a counterexample without "DEFECT"/"FLAW" survives, while "No defect found" blocks. How many probes and defect families?
18. F3 hashes a run key and asserts constant witness properties. F4 emits repair prose for a refused, zero-case run. What does digest-chain checking miss?

Level 7 — Corrections and remaining obligations (Appendix A.5)

19. State necessary evidence conditions in the corrected gate for a declared carrier and for no declared carrier. Does an exact zero alone suffice?
20. A candidate and an accepted checker receipt are digest-bound. Separate byte binding, producer authentication, checker correctness, and critique validity.
21. Historical corrected-gate passing counts are cited; this review only inspects source and retained records. Write a defensible result and name the evidence needed to claim a fresh replay.

Level 8 — Reproducible review records (Sections 6–8)

22. Assign review dispositions: (a) displayed arithmetic checks; (b) subtraction is claimed to make the original sum zero; (c) formal/runtime correspondence is absent; (d) a proposed native command was not executed.
23. List essential execution-record fields. Why are sorted keys insufficient for RFC 8785? What if a rate denominator is zero?

24. Repeated potential adjustments leave the residual unchanged. What should the reviewer record? Does a stagnation trigger license changing inputs?

Capstone — Recount, perturb, and report (8 points)

Predict the counts before running `python tools/review_capstone.py` from the package root. Complete one point for each item:

1. Verify the three retained record digests against their bundled provenance.
2. Check ten distinct conditions: two lattices times five interaction values.
3. Reconstruct the four cells and four exact descriptive fractions.
4. Change one refused CHALLENGE to CONCUR in a copy. Predict and recount.
5. Change only the caught flag in a copy. Predict whether cells change.
6. Group all five gate probe records into four defect families.
7. Preserve originals and identify the altered copy as a control, not a new trial.
8. Write source identity, command, observed outcome, scope, and next obligation.

The tool checks custody, condition census, and recount controls. It does not reexecute models, native checkers, or the historical gate. A matching output does not automatically grade the written explanations.

Practice rubric

Each question earns 0–3 points: one for the result, one for calculation or reason, and one for evidence scope or remaining obligation. The 24 questions total 72 points; the capstone adds eight, for 80 possible. Use scores to find material to revisit. No score establishes competence or publication authority.